Rahul Mondal

+91-8768963374 | rahul425wy@gmail.com | github.com/rahulmondal9

Summary

Data Analyst with hands-on experience in Python, SQL, data cleaning, exploratory data analysis, visualization, and machine learning fundamentals. Skilled in transforming raw data into actionable insights through statistical analysis, dashboard reporting, and model evaluation. Passionate about data-driven decision-making and solving business problems with analytical solutions.

Professional Experience

Data Science & Analytics Intern Aug 2024 – Nov 2024
Zidio Development *Remote*

- Automated data-cleaning pipelines in Python and SQL across multiple datasets, improving data processing efficiency by 40% and reducing manual intervention.
- Built interactive Power BI dashboards visualising key business metrics and trends for stakeholders to support data-driven decisions.
- Built an LSTM speech-emotion classifier (7 emotion classes) on RAVDESS, TESS and SAVEE audio using MFCC, ZCR and RMS features; expanded 4,720 clips to 18,880 samples via noise and pitch augmentation.
- Trained a CNN on MNIST handwritten digits, reaching 98.87% evaluation accuracy; also performed EDA and applied regression and classification models to business datasets.

Data Science & Analytics Intern Mar 2024 – May 2024
Next24Tech *Remote*

- Developed a real-time lane detection system with OpenCV and CNNs (TensorFlow), improving detection accuracy by $\sim$20% under varied lighting conditions.
- Integrated the model into an autonomous-driving simulation pipeline, improving lane-tracking stability in edge-case scenarios.
- Authored technical documentation and collaborated with engineering teams to support onboarding of new team members.

Projects

LLM Response Evaluation Pipeline | *Python, sentence-transformers, scikit-learn, NumPy* Dec 2025 | GitHub

- Built an automated pipeline that scores LLM responses on relevance, completeness and hallucination against retrieved RAG context using local embeddings (all-MiniLM-L6-v2) and cosine similarity, with no LLM API calls.
- Evaluation took 0.03–0.05 s and cost $\sim$\$0.00001 per conversation in test runs; sentence-level detection (similarity < 0.55) flagged 40% of sentences in a case containing an unsupported price claim.
- Wrote an ETL script that normalises noisy raw conversation and context logs into clean JSON for batch evaluation.

Credit Card Fraud Detection | *Python, scikit-learn, Pandas, imbalanced-learn* GitHub

- Built a baseline fraud classifier on 284,807 transactions (492 frauds) using Logistic Regression and SMOTE to handle severe class imbalance.

Technical Skills

Languages: Python, SQL, C
ML & Deep Learning: scikit-learn, TensorFlow/Keras, OpenCV, librosa, sentence-transformers; CNN, LSTM, regression, classification
Data & BI: Pandas, NumPy, Matplotlib, Seaborn, Power BI, MS Excel, Jupyter Notebook; EDA, data cleaning, dashboarding
Tools: Git, GitHub

Education

B.Tech, Computer Science Engineering (Data Science) Sep 2022 – Jun 2026
NSHM Knowledge Campus, Durgapur (MAKAUT) | *CGPA: 7.51/10* *West Bengal, India*
Higher Secondary (West Bengal Board) 2018 – 2020
Baital Gopeswar Pal Vidyapith | *86%* *West Bengal, India*

Certifications & Additional

- Deloitte Data Analytics Job Simulation (Forage), Sep 2026 | practical tasks in data analysis and forensic technology
- TCS Data Visualisation Job Simulation (Forage), Jan 2026
- Effective Writing, NPTEL (IIT Roorkee), Jan–Feb 2023
- Languages: Bengali (fluent), Hindi (fluent), English (professional proficiency)